

CANDIDATE
NAME

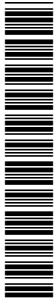
CENTRE
NUMBER

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SCIENCE

Paper 2

1113/02

October 2019

45 minutes

Candidates answer on the Question Paper.

Additional Materials: Pen Calculator
Pencil
Ruler

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You should show all your working in the booklet.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 50.

This document consists of **15** printed pages and **1** blank page.

- 1 The diagram shows the process of photosynthesis in a leaf of a plant.

Complete the diagram.

Choose words from the list.

carbon dioxide

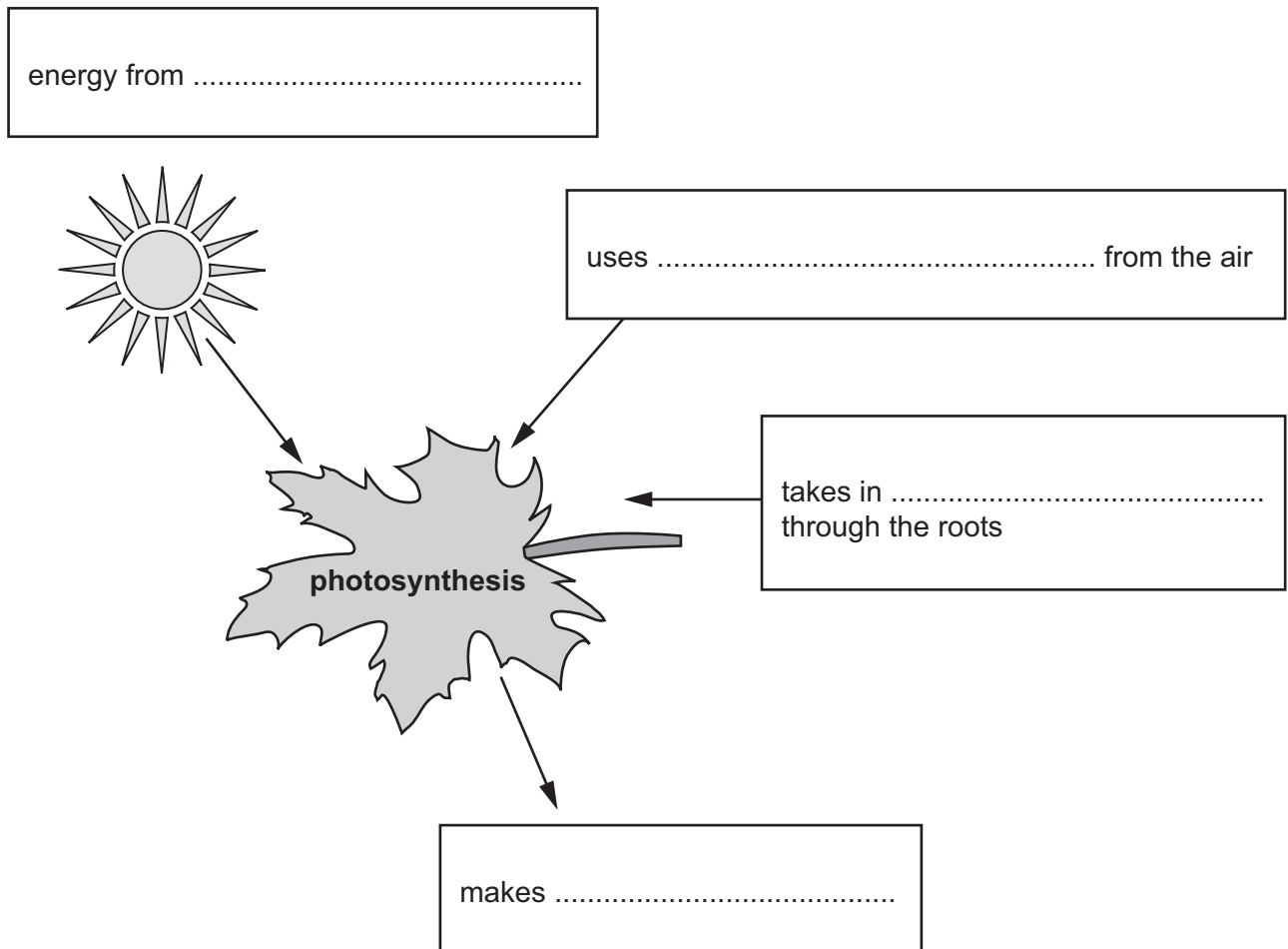
glucose

minerals

nitrogen

sunlight

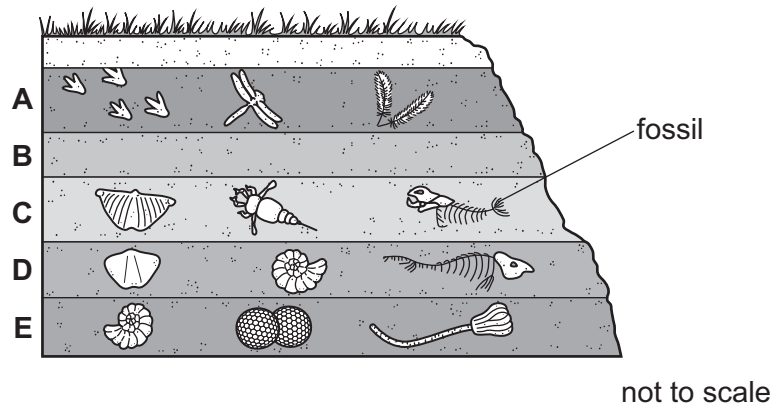
water



[2]

2 Sedimentary rocks are formed in layers.

Look at the diagram of layers of sedimentary rocks.



(a) Which layer has the **youngest** fossils?

.....

[1]

(b) Metamorphic rocks do not normally contain fossils.

Tick (✓) the box next to the correct explanation.

Animals did not live in areas where metamorphic rocks were formed.

☐

Metamorphic rocks are formed when molten rock cools.

☐

Metamorphic rocks were formed before there was life on Earth.

☐

Metamorphic rocks were formed under high temperatures and pressures.

☐

[1]

3 Carlos calculates the density of an object.

(a) The object has a mass of 450 g.

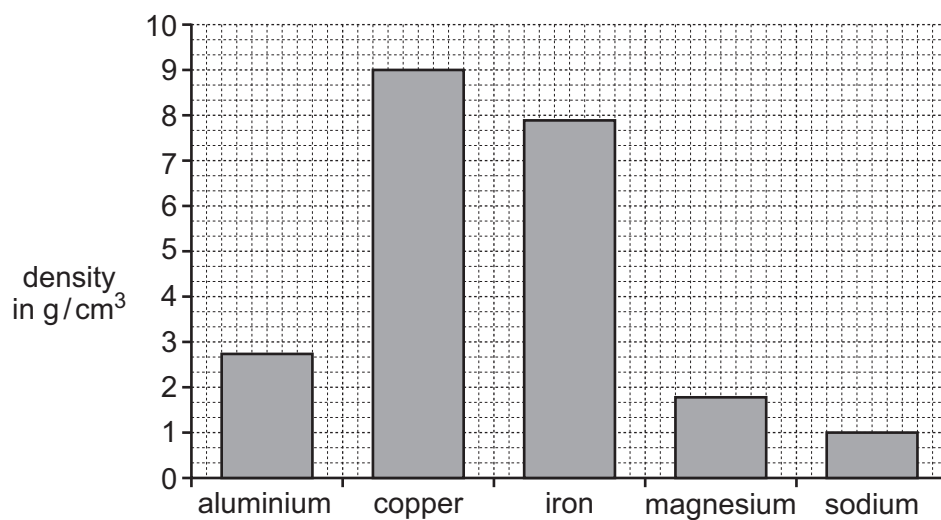
The object has a volume of 50 cm³.

Calculate the density of the object.

density g/cm³

[3]

(b) Carlos knows the densities of different metals.



Carlos has a piece of each of the five metals.

Each piece of metal has the same volume.

Which piece of metal has the **greatest** mass?

Circle the correct answer.

aluminium

copper

iron

magnesium

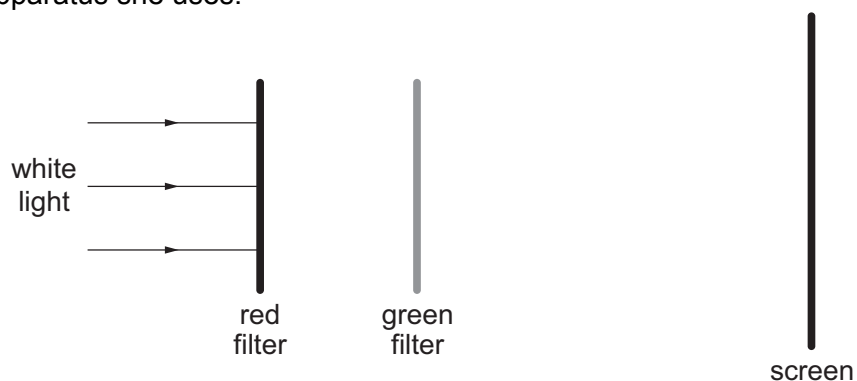
sodium

[1]

4 Jamila does an experiment with light.

She does the experiment in a dark room.

Here is the apparatus she uses.



What does she see on the screen?

.....

Explain your answer.

.....

.....

.....

[3]

5 This question is about physical changes during adolescence.

Complete the table by describing one **physical** change that occurs during adolescence

- **only** in males
- **only** in females
- in **both** males **and** females.

	example of a physical change during adolescence
only in males	
only in females	
in both males and females	

[3]

- 6 A class of 20 students measure their body temperatures.

The thermometer shows the body temperature of student number 20.



- (a) The table shows the results for 19 of the students.

The result for student 20 is missing.

What is the body temperature of this student?

Write your answer in the table.

student	1	2	3	4	5	6	7	8	9	10
temperature in °C	36.9	37.1	37.0	37.1	36.9	37.0	37.1	37.0	37.2	37.1

student	11	12	13	14	15	16	17	18	19	20
temperature in °C	37.0	36.9	37.0	37.2	36.9	37.0	37.1	37.3	37.0	

[1]

- (b) Complete the tally box for all 20 students by counting the number of students at each temperature.

Some of the rows have been done for you.

temperature in °C	tally	number of students at this temperature
36.6		
36.7		
36.8		
36.9		
37.0	HHH II	7
37.1	HHH	5
37.2	II	2
37.3	I	1

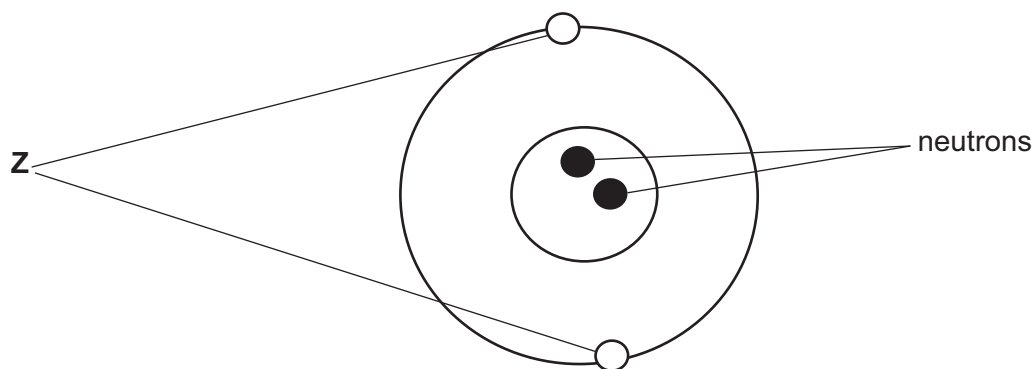
[2]

- (c)** Use the grid to present the results from the completed tally box in part **(b)**.

[illegible]

[3]

- 7 The diagram shows part of the structure of an atom of helium.



- (a) There are three types of particle in an atom.

This diagram shows only two of the types of particle.

- (i) Name the particles labelled **Z**.

..... [1]

- (ii) Name the particles that are missing from the diagram.

.....

Draw on the diagram the correct position and number of the missing particles.

[2]

- (b) The diagram shows the group of the Periodic Table that contains helium.

The elements are in the same order as they appear in the Periodic Table.

helium
neon
argon
krypton
xenon
radon

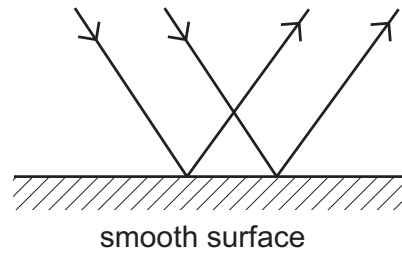
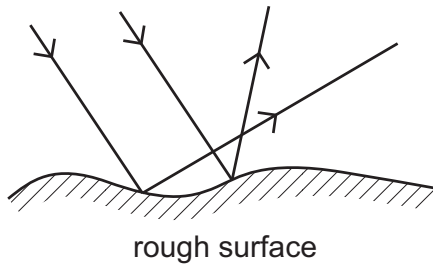
- (i) Which element in the group has the **largest** atomic number?

..... [1]

- (ii) Describe how the radius of the atom changes as you go down the group.

..... [1]

8 Light is reflected from different surfaces.



(a) What is the law of reflection?

Complete the sentence.

The angle of reflection equals the angle of [1]

(b) The reflections on **rough** and **smooth** surfaces are different.

Write down one difference.

.....

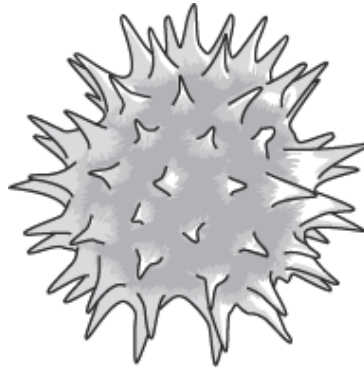
.....

.....

..... [1]

9 Many plants depend on insects such as bees for reproduction.

(a) Look at the diagram of a pollen grain viewed with a microscope.



(i) Name the part of a flower that makes pollen grains.

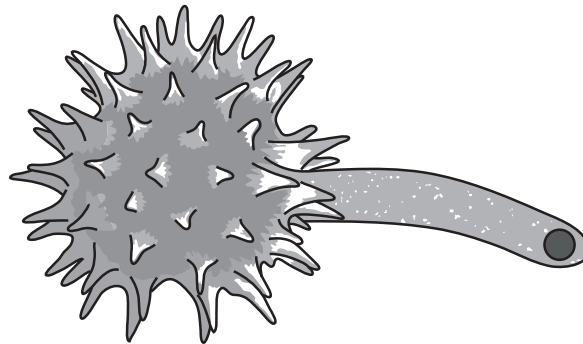
..... [1]

(ii) This pollen grain is adapted to be carried by insects.

Use the diagram to suggest how it is adapted.

.....
 [1]

(b) The following diagram shows the same pollen grain after it has landed on the stigma of a flower.



Explain what is happening to the pollen grain and why this is necessary for reproduction to be completed.

What is happening to the pollen grain?

.....

Why is this necessary?

.....

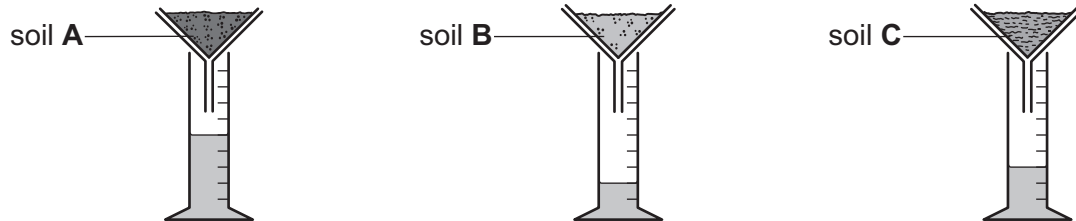
[2]

10 Mia investigates three types of soil, **A**, **B** and **C**.

She wants to find out how much water each soil can hold.

She adds 20 cm³ of water to the same volume of each soil.

She collects the water that comes through in a measuring cylinder.



Look at the results of Mia's investigation.

(a) Match the **letter** of the soil to the correct **name** of the soil.

letter	name of soil
A	clay soil
B	loam soil
C	sandy soil

[2]

(b) Loam soil contains small pieces of decaying plants.

What name describes these pieces of decaying plants?

Choose from the list.

deposits

humus

sediment

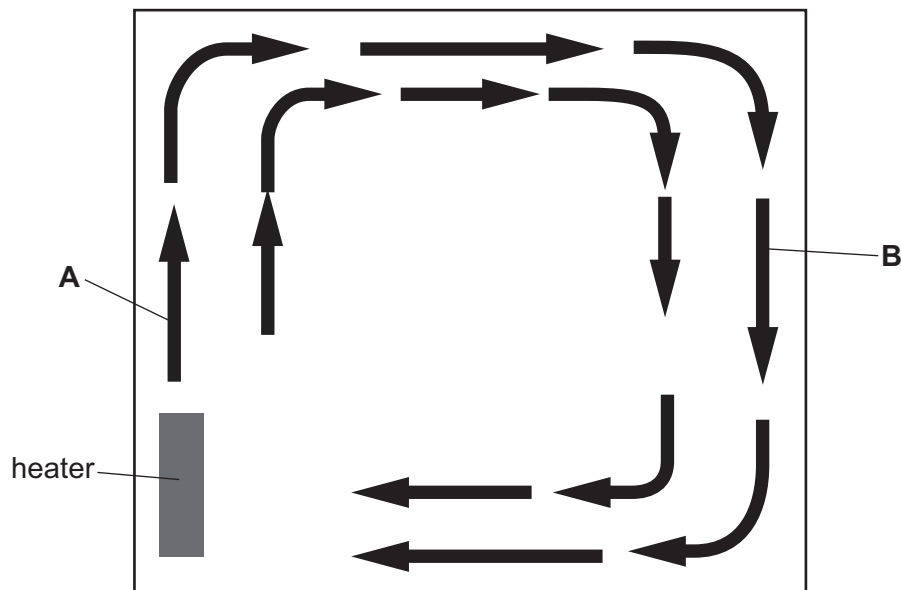
top soil

.....

[1]

11 The diagram shows how a room can be heated.

The arrows show the direction of air movement.



Use the diagram to complete the sentences.

At **A** warm air rises because it is

At **B** cool air sinks because it is

This process is called

[3]

12 Pierre reacts metal compounds with acids to make three different salts.

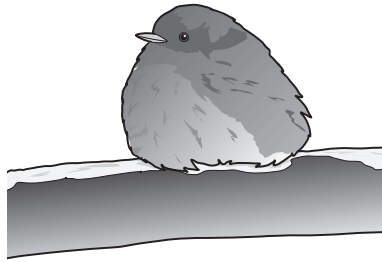
The table shows the metal compounds and the salts he makes.

Complete the table to show the acids he uses in each reaction.

metal compound	acid	salt formed
calcium carbonate		calcium nitrate
sodium hydroxide		sodium sulfate
potassium hydroxide		potassium chloride

[2]

13 Birds can live in cold places.



They trap air between their feathers.

Why does this help to keep the birds warm?

Circle the correct answer.

trapped air is a good conductor

trapped air is a good convector

trapped air is a good insulator

trapped air is a good radiator

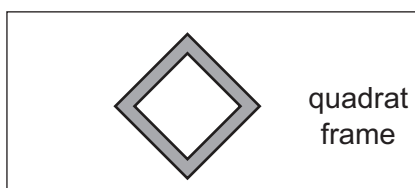
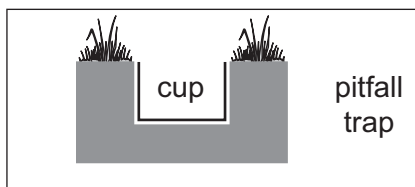
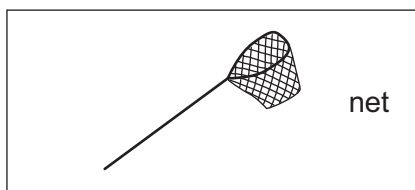
[1]

14 A group of students are doing field work near to their school.

They choose the correct apparatus for some activities.

Draw lines to match each piece of **apparatus** to the **activity**.

apparatus



activity

collecting ground dwelling animals which are active and hunt at night

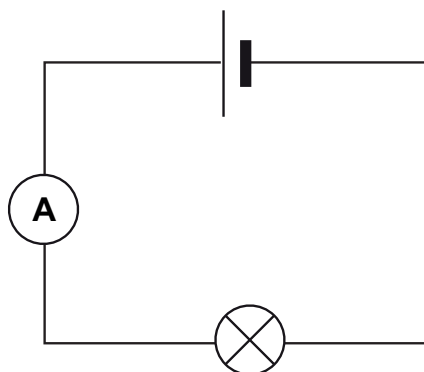
sampling aquatic insects living in a small stream

sampling the plants growing in an area of grassland

[2]

15 Aiko wants to measure different quantities in electrical circuits.

(a) She connects this electrical circuit.



Complete the sentences.

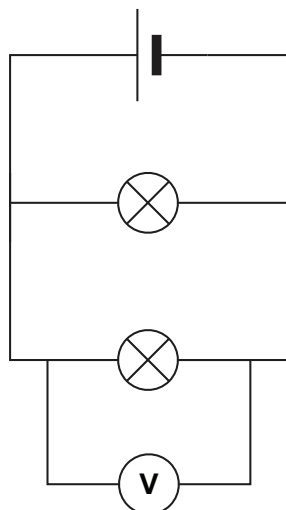
The circuit diagram shows a circuit.

The component shown by **A** is

This component measures

[2]

(b) She connects this electrical circuit.



Complete the sentences.

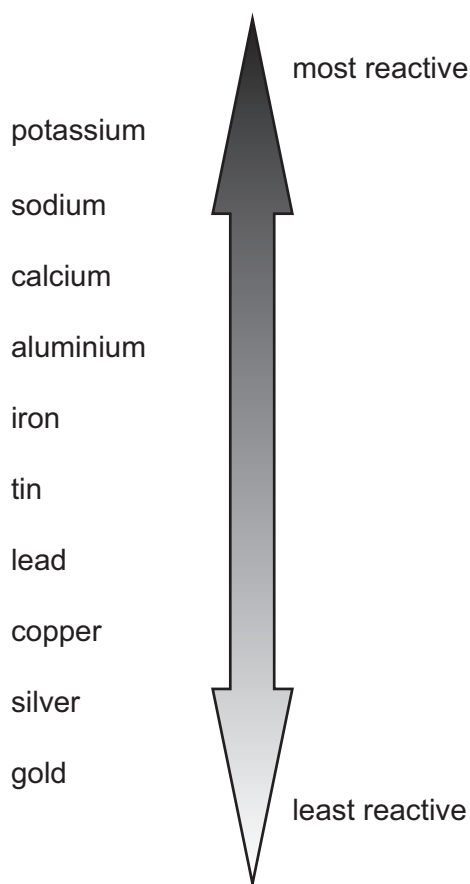
The circuit diagram shows a circuit.

The component shown by **V** is

This component measures the across the lamp.

[2]

16 The diagram shows some of the metals in the reactivity series.



(a) Iron reacts with dilute acid but not cold water.

Write down one metal that reacts with **both** dilute acid **and** cold water.

Choose your answer from the list in the diagram.

..... [1]

(b) Copper reacts with silver nitrate but **not** with lead nitrate.

Explain why.

.....
..... [2]

(c) When copper reacts with silver nitrate two products are made.

Write down the names of these **two** products.

..... and [1]

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